GIT QUEST | 30-DAY GIT LEARNING SERIES

Day 3: The 3 Areas of Git

Week 1: First Steps | Level: Absolute Beginner

Created by Rutuja Naik

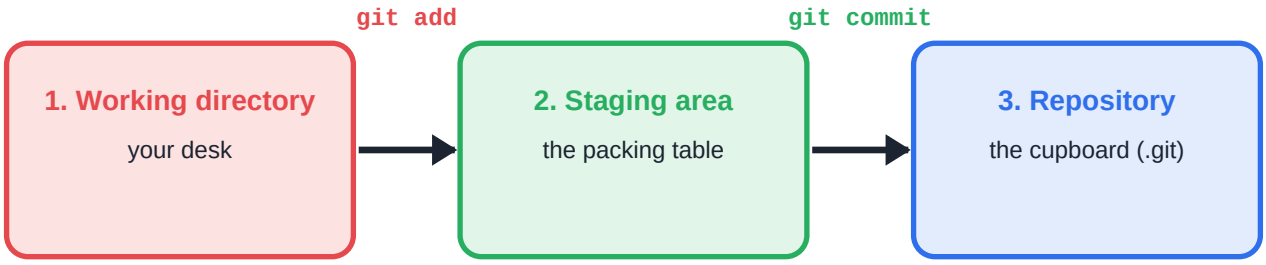
Today's mission

Yesterday you made your first commit. Today you learn **where your files are** while that happens. This one idea makes the rest of Git easy.

- Name the 3 areas: working directory, staging area, repository
- Move files between them with `git add` and `git commit`
- Read `git status` like a traffic light
- Pack only some of your files into a commit

1. The big idea: desk, packing table, cupboard

Imagine sending a parcel. You work on things at your **desk**, pick items and put them on the **packing table**, then seal the box and store it in the **cupboard**. Git works the same way.



Area	Real name	What lives here
1. Your desk	Working directory	Files you are creating and editing right now
2. Packing table	Staging area (the index)	Files you picked for the next save
3. Cupboard	Repository (.git folder)	Sealed save points called commits

Why a packing table at all?
It lets you choose **exactly** what goes into each save point. Change 5 files, commit only 2, and every commit tells one clear story.

2. Step by step

Step 1: Put two new files on your desk

```
$ echo "Milk, eggs" > shopping.txt
$ echo "Project ideas" > notes.txt
```

Step 2: See where they are

```
$ git status
Untracked files:
        notes.txt
        shopping.txt
```

Step 3: Move ONE file to the packing table

```
$ git add shopping.txt
$ git status
Changes to be committed:
        new file:   shopping.txt

Untracked files:
        notes.txt
```

Step 4: Seal it into the cupboard

```
$ git commit -m "Add shopping list"
[main 8c1d4e7] Add shopping list
 1 file changed, 1 insertion(+)
```

Step 5: Check again

```
$ git status
Untracked files:
        notes.txt
```

Step 6: Edit a saved file

```
$ echo "Bread" >> shopping.txt
$ git status
Changes not staged for commit:
        modified:   shopping.txt
```

3. Read git status like a traffic light

Git says	Meaning
Untracked files	RED: a new file Git isn't watching yet (on the desk)
Changes not staged for commit	RED: an edited file, still on the desk
Changes to be committed	GREEN: packed and waiting on the packing table
nothing to commit, working tree clean	BLUE: everything is safely in the cupboard

4. Cheat sheet

Command	What it does
git status	Shows which area each file is in
git add <file>	Desk to packing table
git add .	Pack every changed file at once
git commit -m "message"	Packing table to cupboard
git restore --staged <file>	Take a file back off the packing table

5. Today's 3 commits

#	File	Content	Commit message
1	day03-three-areas/README.md	Today's lesson with steps and cheat sheet	docs(day03): add lesson on the 3 areas of git
2	day03-three-areas/demo.tape	VHS script that records the terminal GIF	feat(day03): add VHS tape for staging demo
3	exercise.md and main README.md	Challenge, and the tracker moves to 3/30	chore(day03): add exercise and update progress tracker

6. Mermaid diagram GitHub draws for you

````
```mermaid
flowchart LR
  A[Working directory] -- git add --> B[Staging area]
  B -- git commit --> C[Repository]
```
````

7. Challenge and quiz

Challenge: edit `shopping.txt`, run `git add`, then `git restore --staged shopping.txt`. Run `git status`: where did the file go?

Q1. Which command moves a file from the desk to the packing table?

Q2. What does "Changes to be committed" mean?

Q3. Where do commits live?

Answers: 1) git add. 2) The file is staged and will go into the next commit. 3) In the repository (.git).

> **Coming up: Day 4**
> **git add and writing good commit messages.** Learn to write messages your future self will thank you for.